CS&IT

Compiler Design

DPP: 2

Parsers

Q1 Consider the following CFG:
S → 1SP2 | 2SQ3
P → 5AB | ε
Q → 6BC | ε
R → 1C | 4
In above grammar S is a start variable, P, Q and R are non terminals, {1, 2, 3, 4, 5, 6} are terminals.
What is in the Follow(S)?
(A) {1, 2, 3, 5, $}
(B) {1, 3, 5, 6, $}
(C) {2, 3, 6, 5, $}
(D) {1, 2, 4, 6, $}

Q2 Consider the following CFG:
S → aSAb | bSBc
A → A+aB | ε
A → B*bC | ε
B → aC | d | ε
What is in the First(A)?
(A) {+, ε, a, d, *}
(B) {ε}
(C) {ε, +, a, d}
(D) {+, ε }

Q3 Which of the following is/are CORRECT statement?
(A) Every left factored grammar can not be LL(1).
(B) The class of grammars that can be parsed using LR methods is proper subset of the class of grammar that can be parsed by LL method
(C) LR parsing is non-backtracking method
(D) If grammar is LL(1) then it will be LR(1) also.

Q4 Consider the following two grammars
G1: S → S1 | 0S1 | 1
G2: A → A0| 1
Which of the following is True regarding above grammars?
(A) G1 is not LR(1)
(B) G2 is LL(1)
(C) Both G1 and G2 is LL(1)
(D) G2 is not LL(1)

Q5 Consider the following grammar.
S → aB | aAb
A → bAb | a
B → aB | ε
How many parse trees are possible to generate the string "aab" from the above grammar?_____

Q6 Consider the augmented grammar given below
S'→S
S→ bP | a
P→ Pa | S
Let I_o = CLOSURE({[S'→.S]}). The number of items in the set GOTO(I_0, b) is:_____

Q7 Consider the following grammar.
S → aSB| c
B → b
The number of reduction steps taken by a bottom-up parser while accepting the string "aaacbbb" is _____.

Q8 Consider the following statements:
1. Canonical LR is more powerful than LL parser.
2. Every SLR(1) grammar is also an LALR(1) grammar.
3. Every LL(1) grammar is also an CLR(1) grammar.
The number of incorrect statements is/are____.

Q9 Consider the given grammar G:
A → aA | Aab | b
Select the correct option.
(A) Given grammar is SLR(1) but not LR(0).
(B) Given grammar is LALR(1) but not SLR(1).
(C) Given grammar is CLR(1) but not LALR(1).
(D) None of these

Q10 Consider the given below statements.
S1: Every CLR(1) grammar is unambiguous.
S2: Every unambiguous grammar is CLR(1)
Select the correct statement:
(A) Both S1 and S2 are correct.
(B) Both S1 and S2 are incorrect.
(C) S1 is correct while S2 is incorrect.
(D) S1 is incorrect while S2 is correct.

Q11 Consider the following statements

S1: If a context free grammar is unambiguous then the language generated by this grammar must be DCFL.
S2: Every CFG corresponding to a DCFL must be unambiguous.
Select the correct option.
(A) Both S1 and S2 are true.
(B) S1 is true and S2 is false.
(C) S1 is false and S2 is true.
(D) Both S1 and S2 are false.

Q12 Consider an augmented grammar
E'→E
E→E+T | T
T→T*F | F
F→(E) | id
If I is the set of one item {[E'→.E]}, then number of items (including I) in CLOSURE(I) are
_____

Answer Key

Q1	(C)
Q2	(A)
Q3	(A, C, D)
Q4	(A, D)
Q5	1~1
Q6	5~5
Q7	7~7
Q8	0~0
Q9	(D)
Q10	(C)
Q11	(D)
Q12	7~7

Hints & Solutions

Q1 Text Solution:

Follow (S) = First (P) ∪ First (Q)

= {5, 2} ∪ {6, 3}

= {2, 3, 5, 6, $}

Option (c) is true.

Q2 Text Solution:

First (A) = First (B) ∪ {+, ε}

= {a, d, *} ∪ {+, ε}

= {a, d, +, *, ε}

Option (a) is correct.

Q3 Text Solution:

(A) Every left factored grammar can not be LL(1). → **True**

(B) The class of grammars that can be parsed using LR methods is proper subset of the class of grammar that can be parsed by LL method. → **False**

(C) LR parsing is non-backtracking method. → **True**

(D) If grammar is LL(1) then it will be LR(1) also. → **True**

Q4 Text Solution:

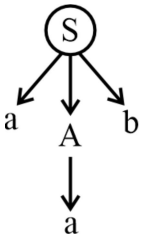
G1: S → S1 | 0S1 | 1

Left recursion. So, not LL(1) and grammar is ambiguos. So, Not CLR (1).

G2: A → A0| 1

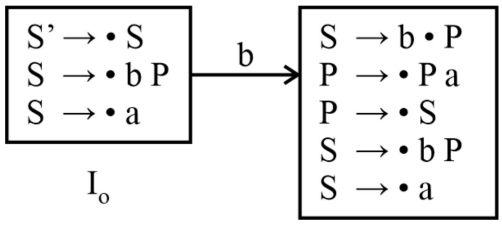
Left recursion. So, not LL(1).

Q5 Text Solution:



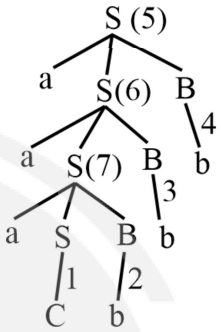
Only one parse tree possible.

Q6 Text Solution:



Number items = 5

Q7 Text Solution:



Total 7 reductions.

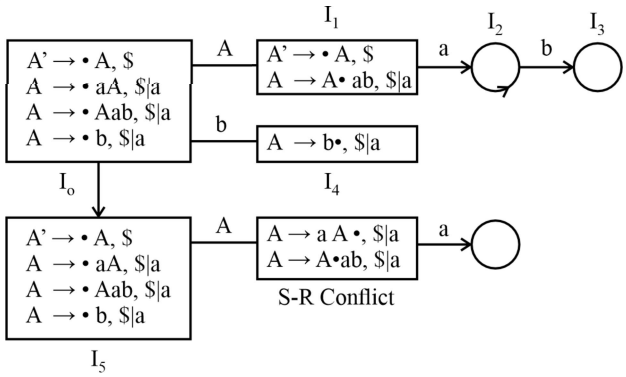
Q8 Text Solution:

1. Canonical LR is more powerful than LL parser. → **True**

2. Every SLR(1) grammar is also an LALR(1) grammar. → **True**

3. Every LL(1) grammar is also an CLR(1) grammar. → **True**

Q9 Text Solution:



Not LR(1) means Not LALR, SLR and LR(0).

Q10 Text Solution:

S1: Every CLR(1) grammar is unambiguous. → **True**
S2: Every unambiguous grammar is CLR(1).
→ **False**

Q11 **Text Solution:**

S1: If a context free grammar is unambiguous then the language generated by this grammar must be DCFL. → **False**
S2: Every CFG corresponding to a DCFL must be unambiguous. → **False**

Q12 **Text Solution:**

$$\begin{aligned} E' &\rightarrow \bullet E \\ E &\rightarrow \bullet E + T \\ E &\rightarrow \bullet T \\ T &\rightarrow \bullet T * F \\ T &\rightarrow \bullet F \\ F &\rightarrow \bullet (E) \\ F &\rightarrow \bullet id \end{aligned}$$

I_o

Number of items = 7


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